

Central Queensland University

CQU Rockhampton TAFE Consolidation - Stage 2

Section J Building Fabric Report

February 2026



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CQU Rockhampton TAFE Consolidation - Stage 2 Section J Building Fabric Report

Central Queensland University

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Approved by:	Ben Gibbs	16/02/2026	BG

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Executive summary

WSP has been engaged to provide a Section J Building Fabric assessment for the proposed CQU Rockhampton TAFE Consolidation Stage 2 project. This analysis has assessed the performance of the building envelope (fabric and glazing) against the NCC 2022 Section J Part 4 and J5 Deemed to Satisfy (DTS) provisions for the conditioned areas of the development.

This assessment only includes Part J4 and J5 of Section J, which cover the architectural requirements. It is the responsibility of the mechanical, electrical, and hydraulic disciplines to demonstrate compliance with the other parts of Section J.

To comply with the requirements of the NCC 2022 Section J, the following building fabric requirements must be met for each section of the building.

Please note that the Building Certifier has confirmed that Section J only applies to any new building or any new portions being retrofitted. Additionally, Section J only applies to conditioned areas and not to unconditioned areas such as workshops, therefore the only buildings with section J requirements are the:

- New Trades Building
- Building 33
- Building 36

Building Fabric Elements	Thermal Requirements
External Envelope Walls - typical	R2.5 added bulk insulation, with R0.2 thermal breaks at all structural members penetrating the insulation layer. No more than 13% of wall area consisting of metal framing. This achieves an overall R-value of at least R1.4 .
External Envelope Walls - Spandrel	R2.0 added bulk insulation required for all spandrel wall sections. Thermally broken frame is not required. This achieves an overall R-value of at least R0.45 .
Internal Envelope Walls	R1.8 added bulk insulation, with R0.2 thermal breaks at all structural members penetrating the insulation layer. No more than 13% of wall area consisting of metal framing. This achieves an overall R-value of at least R1.4 .
Exposed Floor Slab <i>(Suspended slab floor exposed to outside or unconditioned space below)</i>	R1.8 added bulk insulation required.
Floor <i>(slab on ground)</i>	No insulation required.

Building Fabric Elements	Thermal Requirements
Ceiling / Roof	R2.5 added bulk insulation installed above roof purlins with a roof spacer system, AND With a reflective foil facing down with a low emissivity ($\epsilon \leq 0.05$) and adjacent air space > 25mm AND Light coloured roof (Solar Absorptance < 0.45). Note that roof colour is not currently specified on the drawings. Compliant roof colour must be selected.
Glazing	New Trades Building U-Value ≤ 5.8, and SHGC ≤ 0.37 Building 33 U-Value ≤ 3.9, and SHGC ≤ 0.38 Building 36 U-Value ≤ 5.4, and SHGC ≤ 0.62 Please Note: All glazing values are whole system parameters, including both the effects of frame and glass
Skylight	New Trades Building Skylights must be < 5% of floor area served – the documented design is compliant. U-Value ≤ 3.9, and SHGC ≤ 0.49 Please Note: All glazing values are whole system parameters, including both the effects of frame and glass.
Doors	All entrance doors to conditioned spaces must have an airlock, self-closing door, revolving door or rapid roller door or similar, unless the conditioned space has a floor area less than 50m². Therefore, for the proposed buildings, a door type (or similar) listed above must be implemented in the following external doors: New Trades Building GF Main Entry and Secondary Entry. Building 33 L1 Educational Teaching 1, Educational Teaching 2, Computer Lab, Bookable Classroom and Open Workshop. L2 Open Plan Workstation to Outdoor Balcony and Northern Entry. Building 36 GF Restaurant and Training Kitchen. L1 Circulation from Covered Walkway and Hairdressing.

Refer to the mark-up drawings in Appendix A detailing each of these fabric requirements and where they must be applied.

1 Introduction

1.1 Purpose of Report

This report is to provide a statement pertaining to the building envelope performance requirements necessary for the proposed CQU Rockhampton TAFE Consolidation Stage 2 project. The intent is for new fabric and glazing components of all spaces to meet the compliance requirements of Parts J4 Building Fabric and J5 Building Sealing, Section J Energy Efficiency, Volume One of the National Construction Code (NCC) Series 2022 based on the Deemed-to-Satisfy (DTS) provisions.

1.2 Sources of Information

The following sources of information were used to undertake the assessment:

- Parts J4 & J5, Section J, Volume One Amendment 1 of the NCC Series 2022
- Building Certifier review package provided by Knisco for Buildings 33, 36 and 92 New Trades Building, dated 05/01/2026 and 08/01/2026.
- Architectural drawings as noted in Table 1.1 below.

Table 1.1 Reference Documents

Document Name	Document Series	Sheet Number [REVISION #]	Date
Building 33			
Level 1 Floor Plans	CQU-ROK-033-01-ARC	21-01 [4]	19/12/2025
Level 2 Floor Plans	CQU-ROK-033-02-ARC	21-02 [4]	19/12/2025
Elevations	CQU-ROK-033-00-ARC	30-01 [1], 30-02 [1]	19/12/2025
Demolition Plans	CQU-ROK-033-01-ARC	20-01 [3], 20-02 [3]	19/12/2025
Building 36			
Ground Floor plans	CQU-ROK-036-GN-ARC	21-01 [2]	19/12/2025
Level 1 Floor Plans	CQU-ROK-036-01-ARC	21-02 [2]	19/12/2025
Elevations	CQU-ROK-036-00-ARC	30-01 [1], 30-02 [1]	19/12/2025
Demolition Plans	CQU-ROK-036-01-ARC	20-01 [2], 20-02 [2]	19/12/2025
New Trades Building			
Ground Floor plans	CQU-ROK-092-GN-ARC	21-00 [6]	19/12/2025
Level 1 Floor Plans	CQU-ROK-092-01-ARC	21-10 [6]	19/12/2025
Elevations	CQU-ROK-092-00-ARC	30-01 [2], 30-02 [2]	20/01/2026

1.3 Building Classification and Climate Zone

According to the Building Certifier Report by Knisco, the conditioned areas in these three buildings comprise of:

- **New Trades Building** - Class 5 (Offices), Class 8 (Workshops) and Class 9b (Classrooms)
- **Building 33** - Class 5 (Offices) & Class 8 (Workshops)
- **Building 36** – Class 6

All buildings within the site are situated within **Climate Zone 2**.

1.4 Assessment Method

The building envelope is defined by all building elements (roof and ceiling constructions, wall-glazing constructions, and floors) that separate conditioned spaces from external and non-conditioned spaces.

Assessment of Section J, Part J4 requires the proposed fabric of the building envelope (both opaque and transparent elements) to comply with the Deemed-To-Satisfy (DTS) provisions of:

- J4D3 Thermal Construction – general
- J4D4 Roof and ceiling construction
- J4D5 Roof lights
- J4D6 Walls and glazing
- J4D7 Floors

Section J, Part J5 requires the appropriate sealing of relevant building elements to restrict air infiltration.

2 Part J4 Building Fabric

2.1 Walls, Roofs and Floors

Table 2.1 summarises the Part J4 DTS provisions for the thermal performance of building walls, roofs, and floors specific to their building level and orientation.

Insulation R-values are defined in units of m². K/W. The R-values stated in the middle column of Table 2.1 relate to the total system R-values across a built-up construction, while the right column provides the requirements for the insulation specifically.

The Construction R-values below, will be required to demonstrate compliance with the requirements of Part J4. The added insulation requirements below will generally achieve the Construction R-value for a typical metal-framed wall.

Refer to Appendix A for mark-ups showing the extent and locations of the building fabric requirements for the proposed project. Table 2.1 Summary of Part J4 DTS provisions

Building Fabric Elements	Minimum construction R-value requirements including thermal bridging	Added insulation R-values
External Envelope Walls - Typical	R1.4 total construction value (including thermal bridging)	R2.5 added bulk insulation, with R0.2 thermal breaks at all structural members penetrating the insulation layer. No more than 13% of wall area consisting of metal framing. This achieves an overall R-value of at least R1.4 .
External Envelope Walls - Spandrel	R0.45 total construction value (including thermal bridging)	R2.0 added bulk insulation required for all spandrel wall sections. Thermally broken frame is not required. This achieves an overall R-value of at least R0.45 .
Internal Envelope Walls (between conditioned and unconditioned spaces)	R1.4 total construction value (including thermal bridging)	R1.8 added bulk insulation, with R0.2 thermal breaks at all structural members penetrating the insulation layer. No more than 13% of wall area consisting of metal framing. This achieves an overall R-value of at least R1.4 .
Floor (Suspended slab floor exposed to outside or unconditioned space below)	R2.0 total construction value	R1.8 added bulk insulation required.
Floor (slab on ground)	R2.0 total construction value	No insulation required.

Building Fabric Elements	Minimum construction R-value requirements including thermal bridging	Added insulation R-values
Roof	R3.7 total construction value The solar absorptance of the upper surface of the roof must not exceed solar absorptance (SA) 0.45 .	R2.5 added bulk insulation installed above roof purlins with a roof spacer system, AND With a reflective foil facing down with a low emissivity ($\epsilon \leq 0.05$) and adjacent air space > 25mm AND Light coloured roof (Solar Absorptance < 0.45). Note that roof colour is not currently specified on the drawings. Compliant roof colour must be selected.

* Actual performance will depend on construction and level of thermal bridging. Where insulated walls are proposed, thermal bridging has been assessed based on a wall construction with 13% frame. This is considered typical; equal to 90mm studs at 450mm centres with a single noggin, top and bottom track.

All insulation must maintain its thickness and not be compressed. Compressed insulation will not have the same thermal performance as uncompressed insulation and may invalidate the results outlined in this report.

2.2 Glazing

Table 2.2 summarises the DTS requirements for the glazing systems in the project.

The glazing U-value and SHGC given in the table below are whole of window system values that account for the effect of both frame and glass. Glazing orientation is assessed according to the 4 main ordinal compass direction sectors (N/E/S/W).

Table 2.2 Summary of proposed glazing thermal performance parameters.

Area	Whole of System U-Value (including frame and glass)	Note/s
New Trades Building	External windows U-Value ≤ 5.8 , and SHGC ≤ 0.37 Internal windows U-Value ≤ 5.8 , and no requirement for SHGC	These glazing values are whole system parameters, meaning they include the effects of the frame and the glass.
Building 33	U-Value ≤ 3.9 , and SHGC ≤ 0.38	
Building 36	U-Value ≤ 5.4 , and SHGC ≤ 0.62	
Skylight within New Trades Building	U-Value ≤ 3.9 , and SHGC ≤ 0.49	Skylight size cannot exceed beyond 5% of total floor area. Based upon updated drawings, the combined skylights meet this compliance requirement.

Area	Whole of System U-Value (including frame and glass)	Note/s
		These glazing values are whole system parameters, meaning they include the effects of the frame and the glass.

It should be noted that this is the responsibility of the architect and glazing contractor to specify and certify glazing products that meet the stated U-value and SHGC performance requirements.

3 Part J5 Building Sealing

3.1 Doors and windows

All entrance doors to conditioned spaces must have an airlock, self-closing door, revolving door, or rapid roller door or the like unless the conditioned space has a floor area of less than 50 m².

The following rooms have doors that meet these conditions and are required to have some form of self-closing door as noted above:

- **New Trades Building**
 - GF Main entry
 - GF Secondary Entry
- **Building 33**
 - L1 Educational Teaching 1 and Education Teaching 2
 - L1 Open Workshop
 - L1 Computer Lab
 - L1 Bookable Classroom
 - L2 Open Plan Workstation to Outdoor Balcony
 - L2 Northern Entry
- **Building 36**
 - GF Restaurant
 - GF Training Kitchen
 - L1 Circulation from Covered Walkway
 - L1 Hairdressing

These doors are shown in the markups in Appendix A.

Doors and openable windows between conditioned spaces and unconditioned areas or outside should either comply with AS2047 or alternatively must have a seal to restrict air infiltration. This does not apply to fire doors or roller shutter doors which are exempt. The seal must be a draft protection device for the bottom edge of an external swing door, and for other edges may be a foam or rubber compression strip, fibrous seal or similar.

3.2 General Construction

An exhaust fan must be fitted with a sealing device such as a self-closing damper when serving a conditioned space. This must be documented by the mechanical engineer.

External wall insulation must extend beyond the ceiling line and continue up to the roof or floor above. Insulating only to ceiling line will allow heat to enter the ceiling void and migrate into the internal conditioned space, reducing overall thermal performance.

Ceilings, walls, floors, and any openings such as window frames, door frames, roof light frames etc. must be constructed to minimise leakage when they are part of the building envelope between conditioned spaces and outside or unconditioned spaces. This can be done by either:

- Enclosed by internal lining systems that are close fitting at ceiling, wall, and floor junctions, or
- Sealed at junctions and penetrations with close fitting architrave, skirting or cornice, or expanding foam, rubber compressible strip, caulking or similar.

This does not apply to openings, grilles or similar required for smoke hazard management.

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Appendix A

Building Fabric Markups



**CQU ROCKHAMPTON
TAFE
CONSOLIDATION -
STAGE 2**

554/700 Yaamba Rd, Norman
Gardens QLD 4701

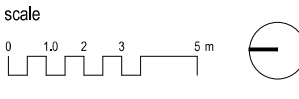
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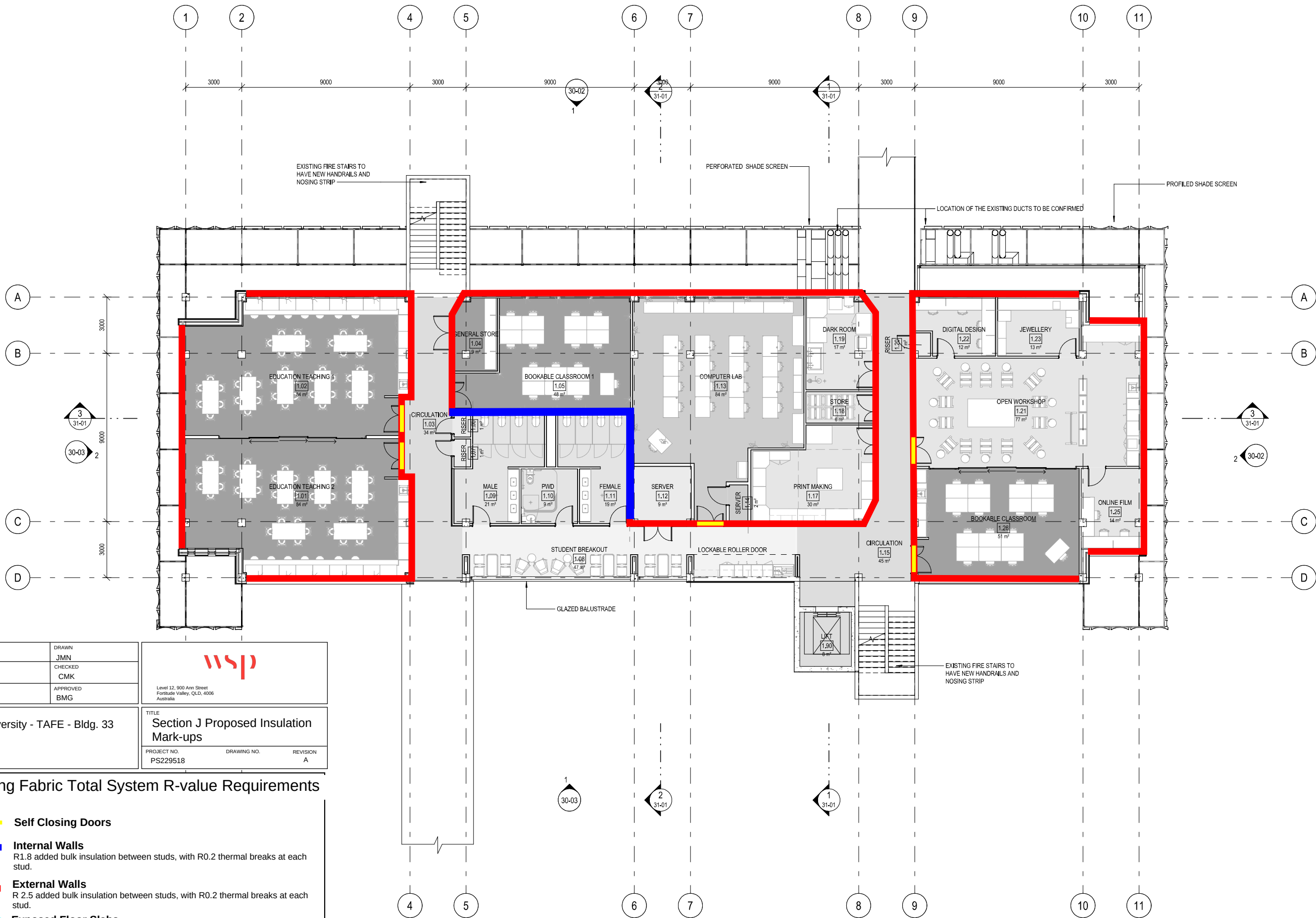
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PROPOSED**

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revision	
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sheet no.	
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PROJECT CQUniversity - TAFE - Bldg. 33		TITLE Section J Proposed Insulation Mark-ups
CLIENT		PROJECT NO. PS229518
		DRAWING NO.
		REVISION A

Building Fabric Total System R-value Requirements

- Self Closing Doors**
- Internal Walls**
R1.8 added bulk insulation between studs, with R0.2 thermal breaks at each stud.
- External Walls**
R 2.5 added bulk insulation between studs, with R0.2 thermal breaks at each stud.
- Exposed Floor Slabs**
R1.8 added bulk insulation
- Ceiling or Roof**
R2.5 added bulk insulation, plus reflective foil face (emissivity <0.05). Solar absorptance (SA) value < 0.45

**CQU ROCKHAMPTON
TAFE
CONSOLIDATION -
STAGE 2**

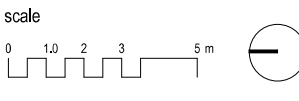
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Gardens QLD 4701

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PT/TA	DEC 2025
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verified	date



**FLOOR PLAN - LEVEL 2 -
PROPOSED**

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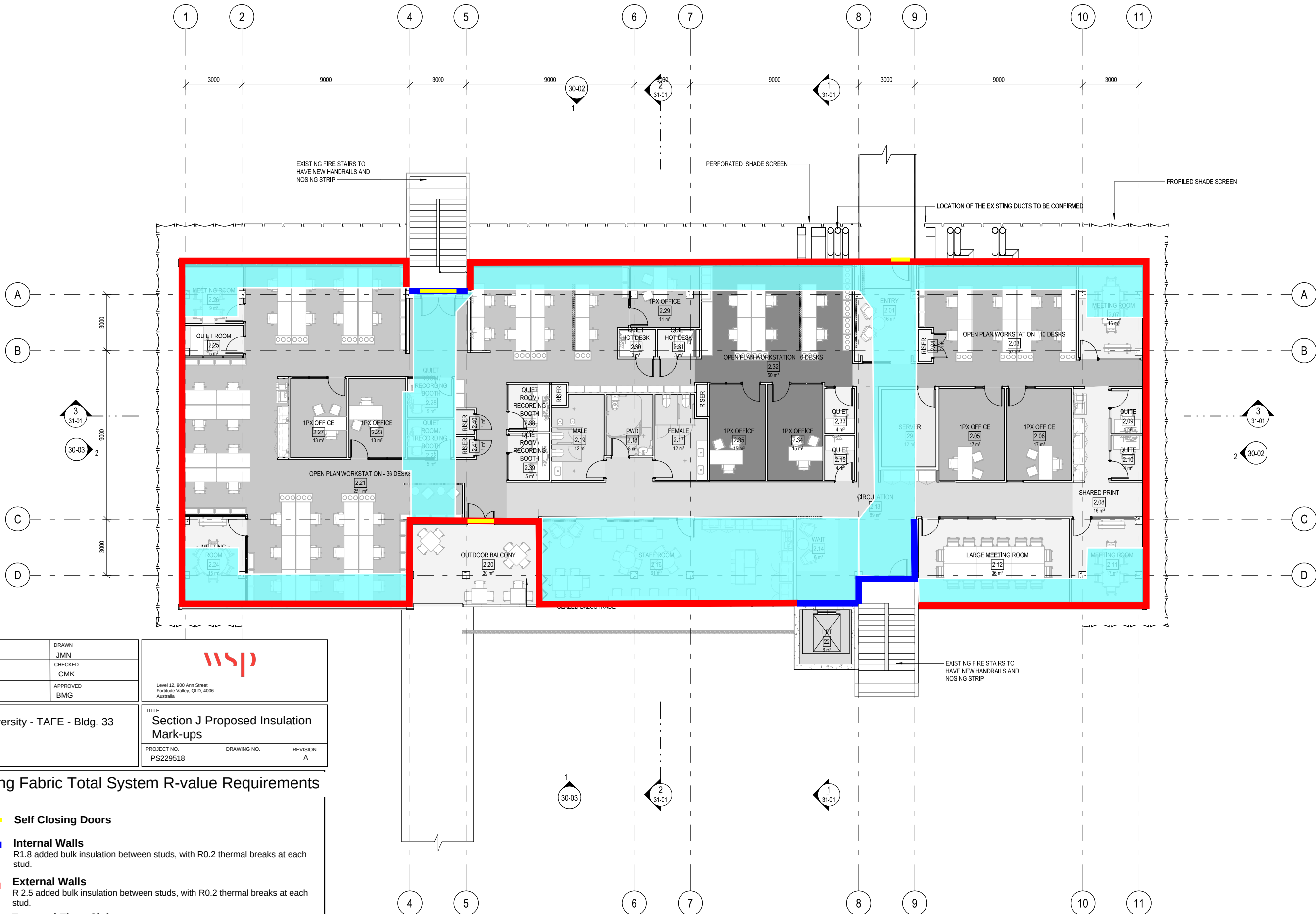
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PROJECT CQUniversity - TAFE - Bldg. 33		TITLE Section J Proposed Insulation Mark-ups	
CLIENT		PROJECT NO. PS229518	REVISION A

Building Fabric Total System R-value Requirements

- Self Closing Doors**
- Internal Walls**
R1.8 added bulk insulation between studs, with R0.2 thermal breaks at each stud.
- External Walls**
R 2.5 added bulk insulation between studs, with R0.2 thermal breaks at each stud.
- Exposed Floor Slabs**
R1.8 added bulk insulation
- Ceiling or Roof**
R2.5 added bulk insulation, plus reflective foil face (emissivity <0.05). Solar absorptance (SA) value < 0.45

**CQU ROCKHAMPTON
TAFE
CONSOLIDATION -
STAGE 2**

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Gardens QLD 4701

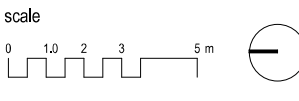
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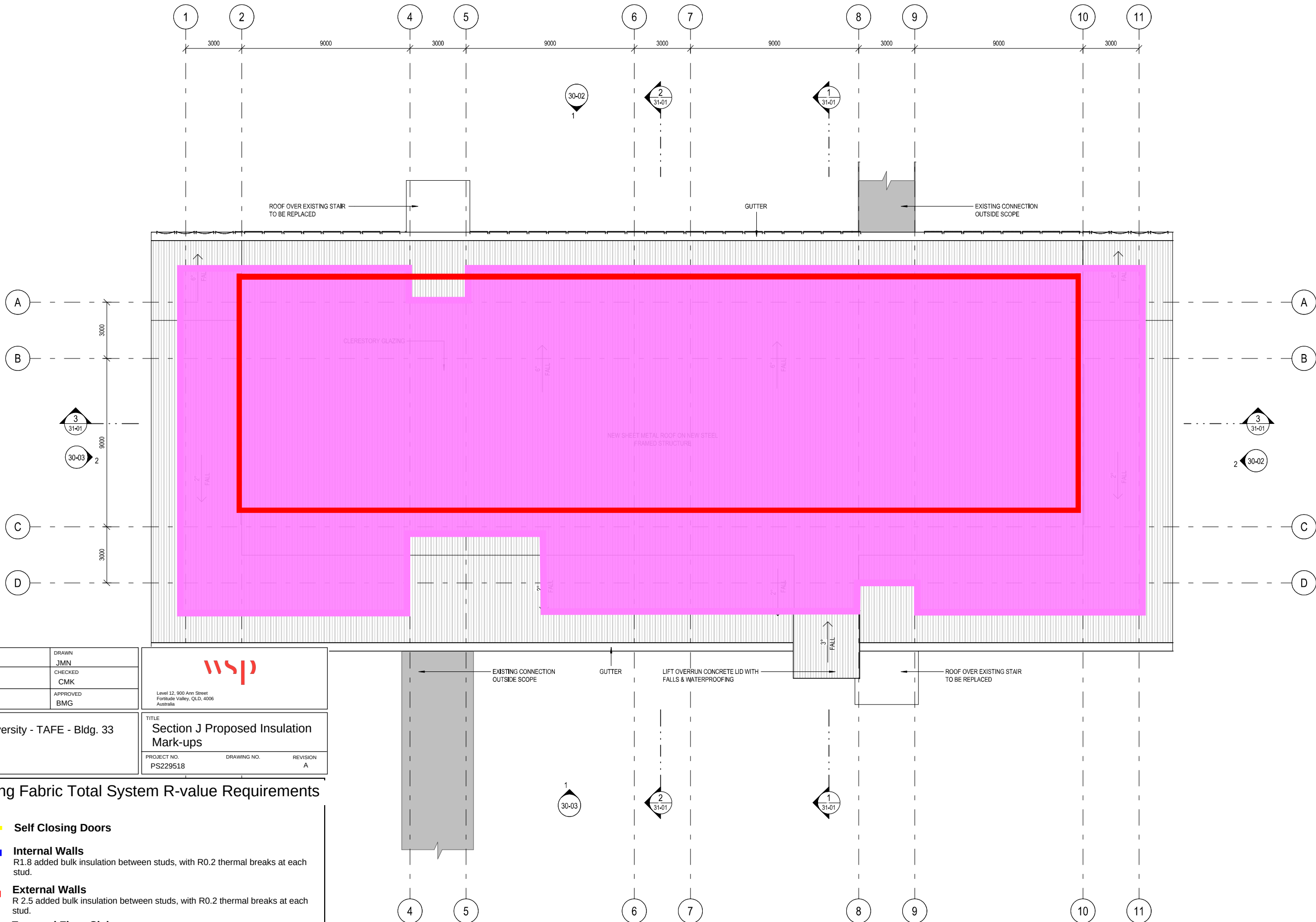
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CLIENT		PROJECT NO. PS229518
		DRAWING NO. A
		REVISION A

Building Fabric Total System R-value Requirements

- Self Closing Doors**
- Internal Walls**
R1.8 added bulk insulation between studs, with R0.2 thermal breaks at each stud.
- External Walls**
R 2.5 added bulk insulation between studs, with R0.2 thermal breaks at each stud.
- Exposed Floor Slabs**
R1.8 added bulk insulation
- Ceiling or Roof**
R2.5 added bulk insulation, plus reflective foil face (emissivity <0.05). Solar absorptance (SA) value < 0.45

**CQU ROCKHAMPTON
TAFE
CONSOLIDATION -
STAGE 2**

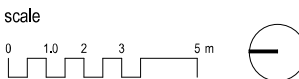
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for



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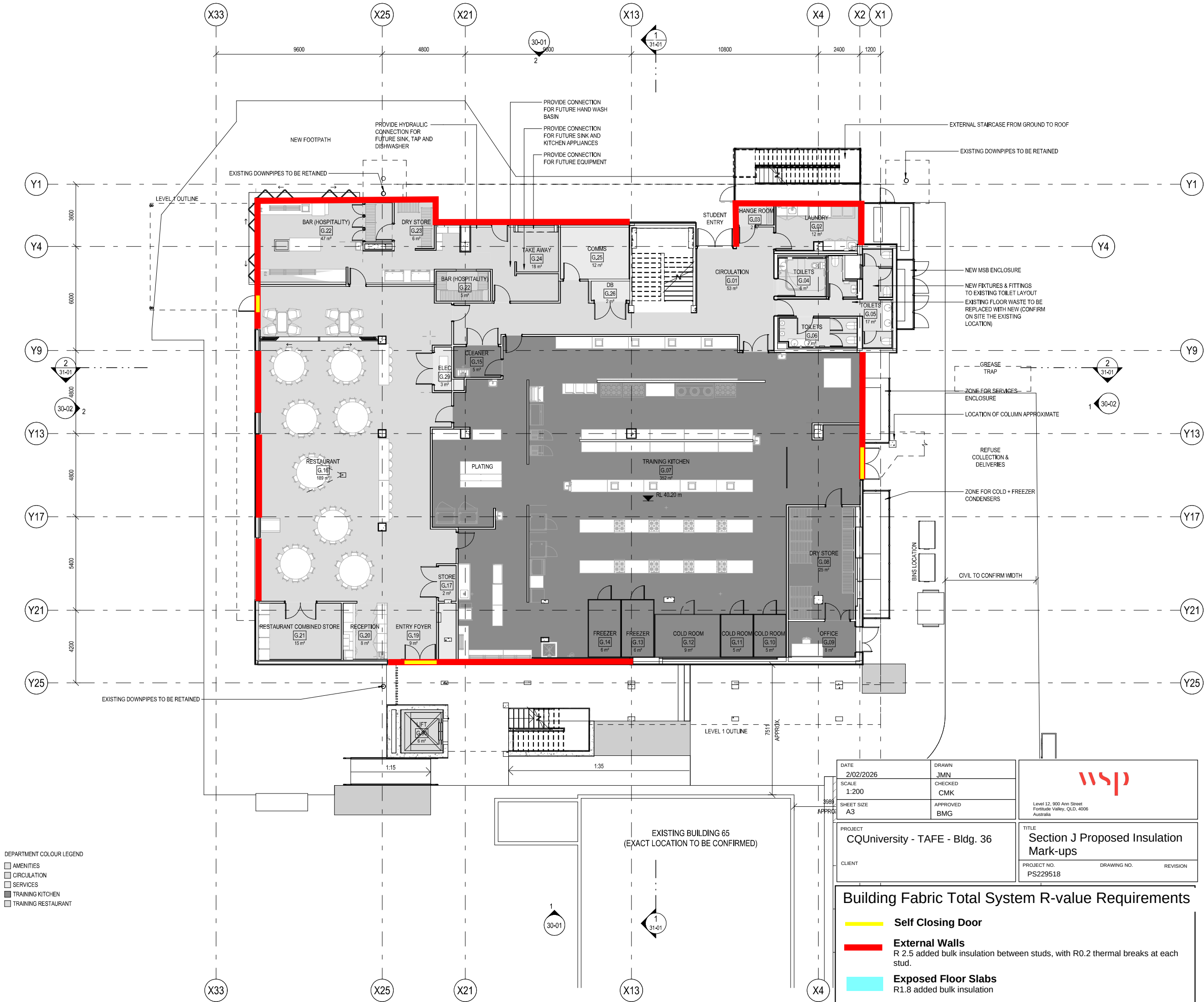


**FLOOR PLAN - GROUND
FLOOR - PROPOSED**

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revision **2**
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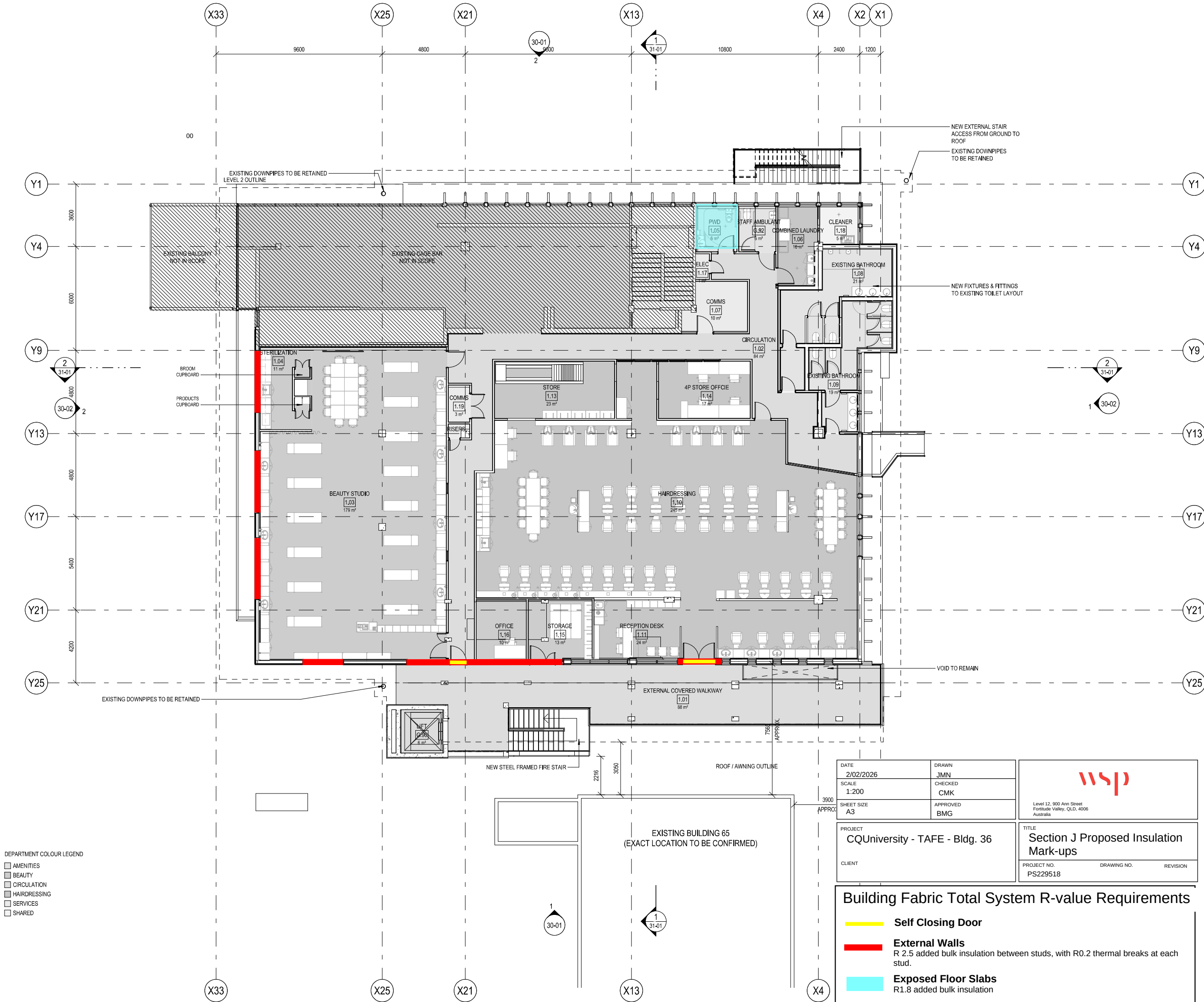


DEPARTMENT COLOUR LEGEND

- AMENITIES
- CIRCULATION
- SERVICES
- TRAINING KITCHEN
- TRAINING RESTAURANT

DATE 2/02/2026	DRAWN JMN	
SCALE 1:200	CHECKED CMK	
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PROJECT CQUniversity - TAFE - Bldg. 36	TITLE Section J Proposed Insulation Mark-ups	
CLIENT	PROJECT NO. PS229518	DRAWING NO. REVISION

Building Fabric Total System R-value Requirements	
	Self Closing Door
	External Walls R 2.5 added bulk insulation between studs, with R0.2 thermal breaks at each stud.
	Exposed Floor Slabs R1.8 added bulk insulation



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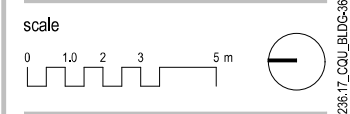
**CQU ROCKHAMPTON
TAFE
CONSOLIDATION -
STAGE 2**

554/700 Yaamba Rd, Norman
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1	2025-12-17	CONSULTANTS REVIEW	BC
rev	date	details	init.

drawn	date
PT/TA	DEC 2025
checked	date
verified	date



**FLOOR PLAN - LEVEL 1 -
PROPOSED**

scale
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project no.
25.0236.17

revision
2

sheet no.
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CLIENT		PROJECT NO. PS229518
		DRAWING NO.
		REVISION

Building Fabric Total System R-value Requirements	
	Self Closing Door
	External Walls R 2.5 added bulk insulation between studs, with R0.2 thermal breaks at each stud.
	Exposed Floor Slabs R1.8 added bulk insulation



DEPARTMENT COLOUR LEGEND

- AMENITIES
- CIRCULATION
- FITTING & MACHINING
- LIGHT AUTOMOTIVE + AUTO ELECTRICS
- METAL FABRICATION
- OUTDOOR
- SERVICES
- SHARED

DATE 2/02/2026	DRAWN JMN	
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SHEET SIZE A3	APPROVED BMG	
PROJECT CQUniversity - TAFE - TRADES Building	TITLE Section J Proposed Insulation Mark-ups	
CLIENT	PROJECT NO. PS229518	DRAWING NO. REVISION A

Building Fabric Total System R-value Requirements

- Self Closing Door**
- Internal Walls**
R1.8 added bulk insulation between studs, with R0.2 thermal breaks at each stud.
- External Walls**
R 2.5 added bulk insulation between studs, with R0.2 thermal breaks at each stud.
- Exposed Floor Slabs**
R1.8 added bulk insulation
- Ceiling or Roof**
R2.5 added bulk insulation, plus reflective foil face (emissivity <0.05). Solar absorptance (SA) value < 0.45
- Skylight**
<5% of the floor area of the space served, ≤ U3.9 and ≤ SHGC0.29

AMENDMENTS

No	Description	Date	By
1	USER GROUP PRESENTATION	12/11/25	SG
2	USER GROUP PRESENTATION	13/11/25	SG
3	USER GROUP PRESENTATION	14/11/25	SG
4	USER GROUP PRESENTATION	18/11/25	SG
5	50% COSTING ISSUE	24/11/25	BS
6	ISSUE FOR COORDINATION	19/12/25	BS



Architect:	Peddle Thorp +61 7 4046 5900
Project Manager:	NA
Structural Engineer:	WSP
Mechanical Engineer:	WSP
Electrical Engineer:	WSP
Hydraulic Engineer:	WSP
Certifier:	KNISCO
Landscape Architect:	TRACT
Civil Engineer:	WSP
Cost Planner:	MBM



PROJECT
CQU ROCKHAMPTON TAFE
CONSOLIDATION - STAGE 2

DRAWING
GA PLAN GROUND LEVEL

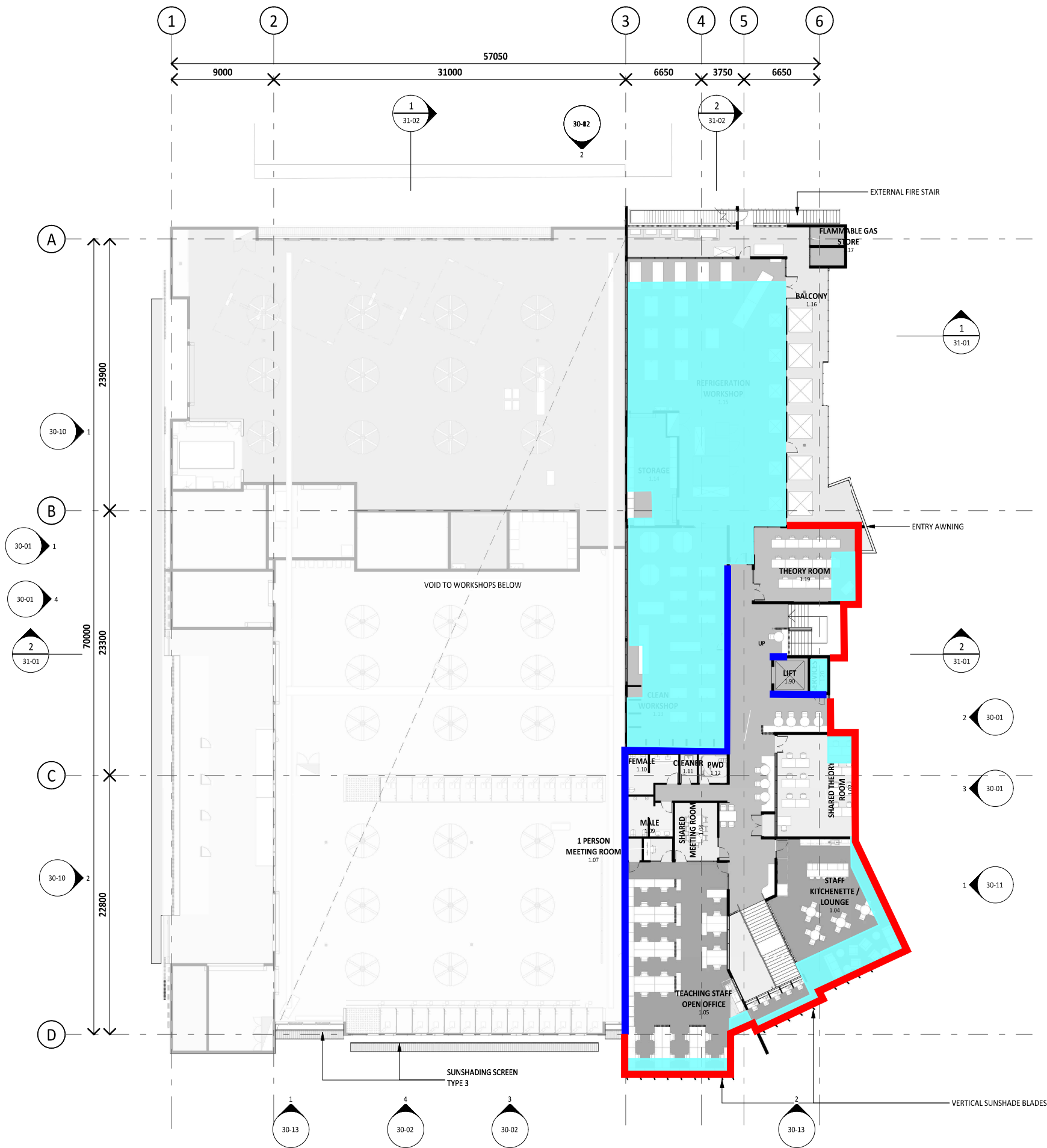
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A1	SCALE (at A1) 1: 200	PROJ. NO. 260018	AUTHORISED BS	DATE 19/12/25
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DRAWING NO. CQU-ROK-092-GN-ARC-21-00	ISSUE 6
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AMENDMENTS			
No	Description	Date	By
1	USER GROUP PRESENTATION	12/11/25	SG
2	USER GROUP PRESENTATION	13/11/25	SG
3	USER GROUP PRESENTATION	14/11/25	SG
4	USER GROUP PRESENTATION	18/11/25	SG
5	50% COSTING ISSUE	24/11/25	BS
6	ISSUE FOR COORDINATION	19/12/25	BS

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PEDDLE THORP
architecture | interiors | health | planning

CONSULTANTS	
Architect:	Peddle Thorp +61 7 4046 5900
Project Manager:	NA
Structural Engineer:	WSP
Mechanical Engineer:	WSP
Electrical Engineer:	WSP
Hydraulic Engineer:	WSP
Certifier:	KNISCO
Landscape Architect:	TRACT
Civil Engineer:	WSP
Cost Planner:	MBM

CLIENT

CQUniversity AUSTRALIA

PROJECT
CQU ROCKHAMPTON TAFE
CONSOLIDATION - STAGE 2

DRAWING
GA PLAN LEVEL 01

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SCALE (at A1)	PROJ. NO.	AUTHORISED	DATE
1: 200	260018	BS	19/12/25

DRAWING NO.	ISSUE
CQU-ROK-092-01-ARC-21-10	6

DATE 2/02/2026	DRAWN JMN	 Level 12, 800 Ann Street Fortitude Valley, Q.L.D. 4006 Australia
SCALE 1:200	CHECKED CMK	
SHEET SIZE A3	APPROVED BMG	
PROJECT CQUniversity - TAFE - TRADES Building		TITLE Section J Proposed Insulation Mark-ups
CLIENT	PROJECT NO. PS229518	REVISION A

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- Skylight**
<5% of the floor area of the space served, ≤ U3.9 and ≤ SHGC0.29

DESIGN DEVELOPMENT

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Fruit	Number of people
Apple	16
Banana	14
Orange	12
Grape	10
Watermelon	8

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DESIGN DEVELOPMENT

Appendix B

Glazing Calculator



NCC 2022 Minimum Glazing Requirements



Project Number	PS229518
Project Name	CQU Tafe - BLDG 33
Location	CQU Rockhampton TAFE
Climate Zone	2
Building Classification	8

Method 1	Glazing		Wall Construction U-value
	U-value	SHGC (External Glazing only)	
North	4.34	0.562	1.40
East	3.56	0.323	1.40
South	3.77	0.297	1.40
West	4.06	0.368	1.40

Method 2	Glazing		Wall Construction R-value
	U-value	SHGC (External Glazing only)	
Whole Building	3.91	0.38	1.40

NCC 2022 Minimum Glazing Requirements



Project Number	PS229518
Project Name	CQU Tafe - BLDG 36
Location	CQU Rockhampton TAFE
Climate Zone	2
Building Classification	8

Method 1	Glazing		Wall Construction U-value
	U-value	SHGC (External Glazing only)	
North	3.80	0.582	1.40
East	5.80	0.810	1.40
South	5.80	0.810	1.40
West	4.21	0.681	1.40

Method 2	Glazing		Wall Construction R-value
	U-value	SHGC (External Glazing only)	
Whole Building	5.44	0.62	1.40

NCC 2022 Minimum Glazing Requirements



Project Number	PS229518
Project Name	CQU Tafe - NEW Trades Building
Location	CQU Rockhampton TAFE
Climate Zone	2
Building Classification	8

Method 1	Glazing		Wall Construction U-value
	U-value	SHGC (External Glazing only)	
North	5.80	0.810	1.40
East	5.80	0.810	1.35
South	5.46	0.340	1.06
West	3.28	0.431	0.81

Method 2	Glazing		Wall Construction R-value
	U-value	SHGC (External Glazing only)	
Whole Building	5.80	0.38	1.10